HOMEWORK 2.5 · UNIT 2 · NC.8.EE.1

Exponent Rules Review

Targets: EE.1-1, EE.1-2, EE.1-3, EE.1-4, EE.1-6, EE.1-7, EE.1-9, EE.1-8

Name _____ Date _____ Block _____

Remember: Multiply same base → **add** exponents: $x^3 \cdot x^4 = x^7$. **Divide** same base → **subtract**: $x^6 \div x^2 = x^4$. **Power to a power** → **multiply**: $(x^2)^3 = x^6$. $x^0 = 1$. $x^{-3} = \frac{1}{x^3}$. Numbers with numbers, letters with letters.

SCORE 3.0 Approaching · the basics

Targets: EE.1-1, EE.1-2, EE.1-3, EE.1-4

1 Find the value. 4^3

REVIEW 2.1 EE.1-1

2 Simplify. $x^6 \cdot x^4$

REVIEW 2.2 EE.1-2

3 Simplify. $(2y^3)^5$

REVIEW 2.3 EE.1-2

4 What is the value of 6^{-2}?

REVIEW 2.3 EE.1-3

Ⓐ $\frac{1}{36}$ Ⓑ -36 Ⓒ -12 Ⓓ 36

5 Simplify. $\frac{30a^5b^7}{5a^4b^2}$

REVIEW 2.2 EE.1-4

SCORE 4.0 Proficient · on target

Targets: EE.1-6, EE.1-7, EE.1-3

6 Simplify. Use positive exponents. $(5m^{-2}n)(2m^{6}n^{-4})$

REVIEW 2.4 EE.1-6

7 **Leo** made a mistake. Circle it. Explain it. Fix it.

REVIEW 2.4 EE.1-7

LEO'S WORK
$(3x^{2})^{3} = 9x^{6}$

8 Which are equal to $\frac{1}{8}$?

EE.1-3

☐ 2^{-3} ☐ 8^{-1} ☐ $(2^{-1})^{3}$ ☐ -8
☐ 4^{-2}

Select all that apply.

SCORE 5.0 Mastery · go deeper

Targets: EE.1-9, EE.1-8

9 Simplify $(x^{4}y^{-2})^{3} \cdot x^{-5}$. Justify each step.

EE.1-9

First _____ . Next _____ . The answer is _____ .

10 Lily says 2^{-4} is a negative number. Is she right? Explain.

EE.1-8

Lily is _____ because 2^{-4} = _____ .